

Myanmar Earthquake & Dam Risk Report

1. Executive Summary

This report covers 9,400 earthquake events in Myanmar from 1970-07-29 to 2026-06-28. The largest event was M7.7 on 1988-11-06. The Gutenberg-Richter b-value is 1.062 with magnitude of completeness $M_c=4.7$.

Total Events: 9,400

Date Range: 1970-07-29 to 2026-06-28

Max Magnitude: M7.7

Mean Depth: 24.8 km

b-value: 1.062

a-value: 7.807

M_c (completeness): 4.7

2. Earthquake Map

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3. Seismic Clustering Analysis

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4. Dam Seismic Risk Assessment

Risk assessment for 254 dams using PGA estimation from the M7.7 mainshock, fault proximity, and structural exposure scoring.

Critical Risk: 25 dams

High Risk: 148 dams

Moderate Risk: 67 dams

Low Risk: 14 dams

Top 30 Highest Risk Dams:

Name	River	PGA(g)	Dist	Fault	Risk	Grade			
Dam near 22.09N 95.86E		0.4019	5.7 km	8.3	Critical				
Dam near 21.77N 95.89E		0.4626	2.7 km	8.3	Critical				
Natkar	0.3593	7.9 km	8.2	Critical					
Dam near 21.47N 95.87E		0.3944	6.0 km	8.2	Critical				
Ta Nai Hka	Ta Nai Hka	0.4441	3.9 km	8.2	Critical				
Dam near 20.72N 95.90E		0.4000	5.6 km	8.2	Critical				
Dam near 21.74N 95.85E		0.3656	7.4 km	8.2	Critical				
Taungnyo	Taungnyo	0.5047	1.2 km	8.2	Critical				
Dam near 20.60N 95.90E		0.3681	7.4 km	8.2	Critical				
Dam near 20.47N 96.08E		0.3252	9.9 km	8.2	Critical				
Dam near 20.85N 96.03E		0.3478	8.3 km	8.2	Critical				
Meikhtila	0.3525	8.2 km	8.2	Critical					
Dam near 17.55N 96.37E		0.3298	9.6 km	8.1	Critical				
Dam near 21.05N 95.83E		0.3105	11.0 km	8.0	Critical				
Dam near 21.04N 95.83E		0.3122	10.9 km	8.0	Critical				
Thaphan Chaung	Thaphan Chaung	0.3041	11.5 km	7.8	Critical				
Dam near 21.34N 95.81E		0.2956	12.2 km	7.7	Critical				
Kun Chaung	Kun Chaung	0.5304	0.9 km	7.6	Critical				
Phyu Chaung	Pyu Chaung	0.5211	1.3 km	7.5	Critical				
Kabaung	Kabaung Chaung	0.3302	9.2 km	7.5	Critical				
Dam near 21.37N 95.80E		0.2830	13.4 km	7.5	Critical				
Yenwe	Ye Nwe Chaung	0.4562	3.9 km	7.4	Critical				
Nyaunggone	0.2795	13.7 km	7.4	Critical					
Dam near 20.73N 95.83E		0.2809	13.6 km	7.4	Critical				
Zeedaw	Nyaung Oat	0.2633	15.3 km	7.2	Critical				
Male-Nattaung	Male-Nattaung	0.2327	18.6 km	6.9	High				
Kyee Ni	0.2362	19.4 km	6.8	High					
Kintat	Mu	0.2235	21.0 km	6.7	High				
Dam near 20.58N 96.17E		0.2281	20.7 km	6.7	High				
Myotha	Ku Hta	0.2211	20.6 km	6.7	High				

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5. Aftershock Forecast

Modified Omori law parameters fitted to aftershocks of M7.7 mainshock.

K (productivity): 4.53

c (time offset): 0.100 hours

p (decay exponent): 0.932

Expected M $\geq$ 3 (7d): 0

Expected M $\geq$ 3 (30d): 0

Expected M $\geq$ 3 (90d): 0

Note: These are statistical estimates based on the Omori decay law. Actual aftershock rates may vary due to local geological conditions.

6. Data Sources

- Earthquake data: mmeq.akze.net API (USGS/ISC sourced)
- Dam data: Open Development Mekong (IFC/WLE), CC BY-SA 4.0
- Fault lines: USGS/Plate boundary data
- PGA estimation: Abrahamson & Silva (2008) NGA-West1 GMPE
- Report generated by mmeq-opendata toolkit